

General Relativity II

Lecture Notes

Metric connections, curvature,
orthonormal frames, and Cartan's equations

Curvature measures the leading change under
parallel transport around an infinitesimal loop.

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1 The metric and covariant differentiation

SOURCE SLIDE 1

GR1 introduced the geometric structures needed to describe fields on a manifold. The metric pairs vectors at the same point, while a connection defines differentiation and parallel transport. We now use the metric to select a connection and study what happens when a vector is transported along different paths. This leads to curvature. We will calculate it both in coordinate bases and in local orthonormal frames.

Throughout, M is a smooth n -dimensional manifold. The notation $C^\infty(M)$ denotes its smooth real functions and $\mathfrak{X}(M)$ its smooth vector fields. In a chart (U, x) , write $\partial_i = \partial/\partial x^i$, with $i, j, k = 1, \dots, n$. In general formulas, coordinate indices range over the dimension of the chart. When specializing to spacetime, we use Greek indices $\mu, \nu, \rho = 0, 1, 2, 3$. Repeated upper and lower indices are summed. All fields and coordinate changes are smooth.

1.1 The metric and its inverse

SOURCE SLIDES 2–3

A metric g assigns to each $p \in M$ a symmetric, nondegenerate bilinear form

$$g_p : T_p M \times T_p M \longrightarrow \mathbb{R}$$

whose components depend smoothly on p . Nondegeneracy means that $g_p(v, w) = 0$ for every $w \in T_p M$ implies $v = 0$. In coordinates,

$$g = g_{ij} dx^i \otimes dx^j, \quad g_{ij} = g(\partial_i, \partial_j), \quad g(v, w) = g_{ij} v^i w^j. \quad (1.1)$$

Thus g_{ij} are the components of the metric tensor, and the matrix (g_{ij}) is invertible at every point.

The signature of a nondegenerate metric is locally constant. We take it to be fixed on the region under consideration. For a Riemannian metric, the quadratic form is positive definite and gives ordinary lengths and angles. For a Lorentzian spacetime metric we use signature $(+, -, -, -)$. A nonzero vector is timelike, null, or spacelike according as $g(v, v)$ is positive, zero, or negative. In particular, $\sqrt{|g(v, v)|}$ is not a norm, since a nonzero null vector has zero value. Euclidean angle formulas require a positive-definite metric, such as $-g$ on an observer's rest space.

The familiar line-element notation abbreviates the symmetric tensor in (1.1),

$$ds^2 = g_{ij} dx^i dx^j.$$

Here $dx^i dx^j$ denotes ordinary multiplication in the quadratic expression. It is not the wedge product $dx^i \wedge dx^j$. For a curve $\gamma : I \rightarrow U$ in this chart, with coordinate functions $x^i(t) = x^i(\gamma(t))$, its meaning is

$$\gamma^* g = g_{ij}(\gamma(t)) \dot{x}^i(t) \dot{x}^j(t) dt \otimes dt.$$

It describes the metric along the curve. A finite coordinate difference between two arbitrary points does not in general give their interval through a single evaluation of g_{ij} .

The inverse matrix defines the inverse metric,

$$g^{ik} g_{kj} = \delta^i_j, \quad g^{-1} = g^{ij} \partial_i \otimes \partial_j. \quad (1.2)$$

It pairs covectors. Raising and lowering indices expresses the inverse maps between tangent and cotangent spaces,

$$v_i = g_{ij}v^j, \quad \alpha^i = g^{ij}\alpha_j.$$

These are metric-dependent operations. They are distinct from the metric-independent pairing $\alpha(v) = \alpha_i v^i$.

For example, Euclidean space has $g_{ij} = \delta_{ij}$ in Cartesian coordinates, while Minkowski spacetime has $g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ in inertial coordinates with $x^0 = ct$. The sphere of radius $a > 0$ has

$$ds^2 = a^2 d\vartheta^2 + a^2 \sin^2 \vartheta d\varphi^2$$

on an angular chart with $0 < \vartheta < \pi$ and a chosen open longitude interval. The component matrices differ because both geometry and choice of coordinates enter their description. Curvature will separate these two effects.

Student work – differentiating the inverse metric

Differentiate $g^{ik}g_{kj} = \delta^i_j$ to show that $\partial_\ell g^{ij} = -g^{im}g^{jn}\partial_\ell g_{mn}$.

1.2 A connection and its action on tensors

SOURCE SLIDES 4–6

Coordinate differentiation of vector components alone does not give a vector. Under a position-dependent change of coordinates, differentiating the component transformation also differentiates the Jacobian. The connection terms compensate for this extra contribution.

An affine connection is a real-bilinear map $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ satisfying

$$\nabla_{fX}Y = f\nabla_XY, \quad \nabla_X(fY) = X(f)Y + f\nabla_XY, \quad f \in C^\infty(M). \quad (1.3)$$

Linearity over smooth functions in the first argument means that the direction at a point is sufficient. The second argument must be a field near that point because it is being differentiated. Define

$$\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k.$$

The first lower index of Γ^k_{ij} specifies the direction of differentiation. Expanding $Y = Y^j\partial_j$ using (1.3) gives

$$\nabla_XY = X^i(\nabla_iY)^k\partial_k, \quad (\nabla_iY)^k = \partial_iY^k + \Gamma^k_{ij}Y^j. \quad (1.4)$$

We also use the conventional component notation ∇_iY^k for $(\nabla_iY)^k$. It includes the connection term and does not mean ordinary differentiation of the scalar component function Y^k .

The connection extends to tensor fields by setting $\nabla_Xf = X(f)$, using the tensor product rule, and requiring differentiation to commute with contractions. For a one-form α , the last requirement gives

$$X(\alpha(Y)) = (\nabla_X\alpha)(Y) + \alpha(\nabla_XY).$$

Taking $X = \partial_i$ and $Y = \partial_j$ gives

$$\partial_i \alpha_j = (\nabla_{\partial_i} \alpha)(\partial_j) + \alpha(\Gamma^k_{ij} \partial_k) = \nabla_i \alpha_j + \Gamma^k_{ij} \alpha_k.$$

Rearranging determines

$$\nabla_i \alpha_j = \partial_i \alpha_j - \Gamma^k_{ij} \alpha_k. \quad (1.5)$$

The minus sign makes the two connection contributions cancel when a vector and a covector are contracted.

Each upper index contributes a positive connection term, and each lower index a negative one. For a tensor of type (r, s) , this gives

$$\begin{aligned} \nabla_k T^{i_1 \dots i_r}_{j_1 \dots j_s} &= \partial_k T^{i_1 \dots i_r}_{j_1 \dots j_s} \\ &+ \sum_{a=1}^r \Gamma^{i_a}_{k\ell} T^{i_1 \dots \ell \dots i_r}_{j_1 \dots j_s} \\ &- \sum_{b=1}^s \Gamma^{\ell}_{kj_b} T^{i_1 \dots i_r}_{j_1 \dots \ell \dots j_s}. \end{aligned} \quad (1.6)$$

Here a and b only count index slots. In each summand, ℓ occupies the indicated slot. For fixed X , $\nabla_X T$ has the same tensor type as T . Retaining the direction as an extra covariant argument gives ∇T , a tensor of type $(r, s+1)$.

The coefficients Γ^k_{ij} themselves are not tensor components. For overlapping coordinates x^i and $x^{i'}$, they obey

$$\Gamma^{k'}_{i'j'} = \frac{\partial x^{k'}}{\partial x^k} \frac{\partial x^i}{\partial x^{i'}} \frac{\partial x^j}{\partial x^{j'}} \Gamma^k_{ij} + \frac{\partial x^{k'}}{\partial x^k} \frac{\partial^2 x^k}{\partial x^{i'} \partial x^{j'}}. \quad (1.7)$$

This follows by differentiating $\partial_{j'} = (\partial x^j / \partial x^{j'}) \partial_j$ with $\nabla_{\partial_{i'}}$. The second derivative is precisely the term absent from a tensor transformation law. Nonzero connection coefficients can therefore occur in flat space.

1.3 Metric compatibility and torsion

SOURCE SLIDES 5–7

To make differentiation respect the metric pairing, require

$$(\nabla_X g)(Y, Z) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z) = 0. \quad (1.8)$$

This condition is *metric compatibility*, written $\nabla g = 0$. It does not require the scalar function $g(Y, Z)$ to be constant for arbitrary fields. Its change is accounted for by the changes in the two fields. In coordinates,

$$0 = \nabla_i g_{jk} = \partial_i g_{jk} - \Gamma^\ell_{ij} g_{\ell k} - \Gamma^\ell_{ik} g_{j\ell}. \quad (1.9)$$

Since covariant differentiation commutes with contraction, $\nabla_i \delta^j_k = 0$. Differentiating (1.2) gives

$$0 = \nabla_i (g^{j\ell} g_{\ell k}) = (\nabla_i g^{j\ell}) g_{\ell k} + g^{j\ell} \nabla_i g_{\ell k} = (\nabla_i g^{j\ell}) g_{\ell k}.$$

Multiplying by g^{km} yields $\nabla_i g^{jm} = 0$. Raising and lowering indices therefore commute with a metric-compatible covariant derivative.

The *torsion* of a connection is the tensor

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]. \quad (1.10)$$

The derivatives of multiplying functions cancel, so this expression is $C^\infty(M)$ -linear in both arguments. Coordinate fields commute, hence

$$T(\partial_i, \partial_j) = T^k_{ij} \partial_k, \quad T^k_{ij} = \Gamma^k_{ij} - \Gamma^k_{ji}.$$

Thus zero torsion is equivalent to symmetry of the lower connection indices in a coordinate basis. In a general frame, the bracket term in (1.10) must be retained.

1.4 The Levi-Civita connection

SOURCE SLIDE 7

Theorem 1.1: The metric determines a torsion-free connection

Every smooth pseudo-Riemannian metric g determines a unique metric-compatible, torsion-free connection. Its coordinate coefficients are

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}). \quad (1.11)$$

This is the *Levi-Civita connection*, and its coefficients are the *Christoffel symbols* in the chosen coordinates.

To obtain the formula, write metric compatibility in three orders,

$$\begin{aligned} \partial_i g_{j\ell} &= \Gamma^m_{ij} g_{m\ell} + \Gamma^m_{i\ell} g_{jm}, \\ \partial_j g_{i\ell} &= \Gamma^m_{ji} g_{m\ell} + \Gamma^m_{j\ell} g_{im}, \\ \partial_\ell g_{ij} &= \Gamma^m_{\ell i} g_{mj} + \Gamma^m_{\ell j} g_{im}. \end{aligned}$$

Add the first two equations and subtract the third. The symmetry $\Gamma^m_{ij} = \Gamma^m_{ji}$ cancels four terms and combines the remaining two,

$$\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij} = 2g_{m\ell} \Gamma^m_{ij}.$$

Multiplication by $g^{k\ell}/2$ gives (1.11). Any connection with the required properties must have these coefficients, proving uniqueness.

For existence, define a connection on each coordinate domain using (1.11) and (1.4). Its coefficients are smooth and symmetric in i, j . Substitution in (1.9) verifies metric compatibility. On overlapping domains the two local connections have the same required properties, so uniqueness makes them agree. They consequently define a connection on all of M . This is the coordinate construction of the connection treated intrinsically in GR1.

Coordinates adapted to one point. The coordinate construction in GR1 can now be summarized directly. Start with coordinates centered at p , and define new coordinates y^i near p by

$$x^i = y^i - \frac{1}{2} \Gamma^i_{jk}(p) y^j y^k.$$

The Jacobian at p is the identity, so this is locally invertible. The second derivatives at p are

$$\left. \frac{\partial^2 x^i}{\partial y^j \partial y^k} \right|_p = -\frac{1}{2} (\Gamma^i_{jk}(p) + \Gamma^i_{kj}(p)) = -\Gamma^i_{jk}(p),$$

where the final equality uses zero torsion. Substitution into (1.7) gives $\Gamma^{i'}_{j'k'}(p) = \Gamma^i_{jk}(p) - \Gamma^i_{jk}(p) = 0$, with old and new index labels identified by the identity Jacobian. For the Levi-Civita connection, metric compatibility then gives $\partial_{i'}g_{j'k'}(p) = 0$. A preceding linear change of coordinates can also put $g(p)$ into its standard diagonal form. These conditions hold at the chosen point. They impose no vanishing condition on derivatives of Γ there, and do not make the coefficients vanish on a neighborhood.

Student work – a conformally Euclidean metric

For $g_{ij} = e^{2f}\delta_{ij}$ on an open subset of $\mathbb{R}^n$, derive

$$\Gamma^k_{ij} = \delta^k_j \partial_i f + \delta^k_i \partial_j f - \delta_{ij} \delta^{k\ell} \partial_\ell f.$$

From now on, unless stated otherwise, ∇ is the Levi-Civita connection. The metric has supplied a definite transport rule. The next question is whether transporting the same vector between two nearby points gives the same result along different paths. Curvature measures the local failure of this independence.

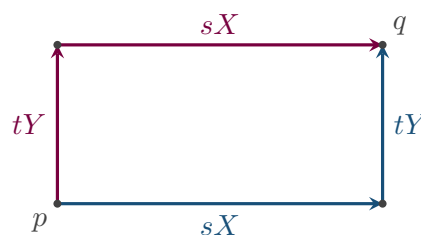
2 Parallel transport and curvature

SOURCE SLIDE 8

A connection specifies parallel transport along a chosen curve. It does not generally make the result independent of the curve. In GR1, transport around a latitude on the sphere returned a vector with a different direction. We now examine this effect around an arbitrarily small loop. The leading change defines curvature locally, without referring to an embedding of the manifold in a larger space.

2.1 Comparing two paths

Let p lie in a coordinate chart and choose two coordinate directions $X = \partial_\mu$ and $Y = \partial_\nu$, with $\mu \neq \nu$. Keep all other coordinates fixed. Starting at p , increasing these two coordinates by small amounts s and t produces a coordinate rectangle. The two orders of travel reach the same point q because coordinate vector fields commute.



Write $P_{XY} : T_p M \rightarrow T_q M$ for transport first along X , then along Y , and P_{YX} for the other route. Given $v \in T_p M$, the vectors $P_{XY}v$ and $P_{YX}v$ belong to the same tangent space and can

be compared. We can also return the first vector to p along the reverse of the second route. The resulting endomorphism

$$H_{s,t} = P_{YX}^{-1} P_{XY} : T_p M \longrightarrow T_p M$$

is the transport map around the oriented loop $+X, +Y, -X, -Y$. A transport map around a closed loop is called a *holonomy map*.

These paths are coordinate segments. They need not be geodesics, and X and Y need not be parallel along them. Using coordinate segments separates the issue of comparing transport from the additional problem of constructing curves with prescribed covariant acceleration.

2.2 The curvature operator

SOURCE SLIDE 9

Successive covariant derivatives provide the local version of this comparison. For general vector fields, the directions themselves need not commute. Their failure to commute must be removed to obtain an expression that depends only on the vectors at the point.

Definition 2.1: Curvature

The curvature of a connection ∇ is the operator

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, \quad X, Y, Z \in \mathfrak{X}(M). \quad (2.1)$$

Our component convention is

$$R(\partial_\mu, \partial_\nu) \partial_\sigma = R^\rho{}_{\sigma\mu\nu} \partial_\rho. \quad (2.2)$$

The final two component indices specify the two differentiation directions. The second index specifies the vector on which the curvature operator acts. In particular,

$$R(X, Y)Z = R^\rho{}_{\sigma\mu\nu} Z^\sigma X^\mu Y^\nu \partial_\rho.$$

Although its definition uses derivatives, curvature is linear over $C^\infty(M)$ in all three arguments. To see the cancellation in the first argument, use $[fX, Y] = f[X, Y] - Y(f)X$. Then

$$\begin{aligned} R(fX, Y)Z &= f \nabla_X \nabla_Y Z - Y(f) \nabla_X Z - f \nabla_Y \nabla_X Z \\ &\quad - f \nabla_{[X, Y]} Z + Y(f) \nabla_X Z \\ &= f R(X, Y)Z. \end{aligned}$$

Skew symmetry in X, Y gives the same property in the second argument. Expanding the third argument by the product rule gives

$$R(X, Y)(fZ) = f R(X, Y)Z + (X(Y(f)) - Y(X(f)) - [X, Y](f))Z.$$

The last coefficient vanishes by the definition of the Lie bracket. Thus R is a tensor field of type $(1, 3)$. At a point p , $R(X, Y)Z$ depends only on X_p, Y_p, Z_p , rather than their chosen extensions away from p .

Student work – curvature as an endomorphism

For fixed $u, v \in T_p M$, show that $R_p(u, v)$ is a linear map $T_p M \rightarrow T_p M$. Verify $R_p(u, v) = -R_p(v, u)$.

2.3 The coordinate formula

For this calculation write $D_\mu = \nabla_{\partial_\mu}$ as an operator on vector fields. The connection coefficients satisfy

$$(D_\mu V)^\rho = \partial_\mu V^\rho + \Gamma^\rho_{\mu\sigma} V^\sigma.$$

Apply this rule once more, keeping the direction ∂_ν fixed in the operator D_ν . It gives

$$\begin{aligned} (D_\mu D_\nu V)^\rho &= \partial_\mu \partial_\nu V^\rho + (\partial_\mu \Gamma^\rho_{\nu\sigma}) V^\sigma + \Gamma^\rho_{\nu\sigma} \partial_\mu V^\sigma \\ &\quad + \Gamma^\rho_{\mu\lambda} \partial_\nu V^\lambda + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} V^\sigma. \end{aligned}$$

Subtract the expression with μ and ν exchanged. The second partial derivatives cancel. The first derivatives of V cancel in pairs after relabelling dummy indices. Since $[\partial_\mu, \partial_\nu] = 0$, the remaining coefficient is

$$\boxed{R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}.} \quad (2.3)$$

The derivatives and products of connection coefficients combine into a tensor even though the individual connection coefficients do not.

There are two related second derivatives in this calculation. The iterated operator $D_\mu D_\nu V$ differentiates a vector field with a fixed coordinate direction in its inner operator. The full covariant second derivative differentiates the vector-valued one-form ∇V . Its direction index is a covariant index, so the tensor differentiation rule contributes an additional negative connection term. Intrinsically,

$$(\nabla^2 V)(X, Y) = \nabla_X \nabla_Y V - \nabla_{\nabla_X Y} V.$$

Its components therefore contain an additional term,

$$(\nabla^2 V)^\rho_{\mu\nu} = (D_\mu D_\nu V)^\rho - \Gamma^\lambda_{\mu\nu} (D_\lambda V)^\rho.$$

Antisymmetrizing therefore gives, for a general connection,

$$(\nabla^2 V)^\rho_{\mu\nu} - (\nabla^2 V)^\rho_{\nu\mu} = R^\rho_{\sigma\mu\nu} V^\sigma - T^\lambda_{\mu\nu} (D_\lambda V)^\rho.$$

The fixed-direction operator commutator has no torsion term. For the Levi-Civita connection, $T = 0$, so both antisymmetrizations give

$$(\nabla^2 V)^\rho_{\mu\nu} - (\nabla^2 V)^\rho_{\nu\mu} = R^\rho_{\sigma\mu\nu} V^\sigma. \quad (2.4)$$

This is the meaning of the usual commutator formula when its successive covariant derivatives are written in component notation.

2.4 Curvature and a small loop

The two routes agree whenever either side length vanishes. Their difference therefore has no term depending only on s or only on t . We will calculate the first possible nonzero term, proportional to st , and then transport the difference back to $T_p M$.

Introduce the matrices $(\Gamma_\mu)^\rho_\sigma = \Gamma^\rho_{\mu\sigma}$, with μ fixed. Along a short segment in the μ direction, the parallel-transport equation gives

$$V(s) = (I - s\Gamma_\mu(p))V(0) + O(s^2),$$

where I is the identity matrix in the coordinate frame. For the second segment we must also expand the connection at its shifted starting point. If p_s is the endpoint of the first segment, then

$$\Gamma_\nu(p_s) = \Gamma_\nu(p) + s\partial_\mu\Gamma_\nu(p) + O(s^2).$$

The first-order segment matrices are therefore multiplied in the order

$$[I - t(\Gamma_\nu + s\partial_\mu\Gamma_\nu)][I - s\Gamma_\mu].$$

The right factor acts first. The product contributes $st\Gamma_\nu\Gamma_\mu$, while the shifted connection contributes $-st\partial_\mu\Gamma_\nu$. All matrices and derivatives in the following expansion are evaluated at p . Along the μ segment, differentiating $V' = -\Gamma_\mu V$ once more gives

$$V'' = -(\partial_\mu\Gamma_\mu)V - \Gamma_\mu V' = (\Gamma_\mu^2 - \partial_\mu\Gamma_\mu)V.$$

This determines the pure quadratic terms in the Taylor expansion. Representing transport in the coordinate bases at the endpoints gives

$$\begin{aligned} P_{XY} &= I - s\Gamma_\mu - t\Gamma_\nu + \frac{s^2}{2}(\Gamma_\mu^2 - \partial_\mu\Gamma_\mu) + \frac{t^2}{2}(\Gamma_\nu^2 - \partial_\nu\Gamma_\nu) \\ &\quad + st(\Gamma_\nu\Gamma_\mu - \partial_\mu\Gamma_\nu) + O((|s| + |t|)^3). \end{aligned}$$

Exchanging the order of the segments leaves the pure terms unchanged and replaces the mixed coefficient by $\Gamma_\mu\Gamma_\nu - \partial_\nu\Gamma_\mu$. Thus

$$\begin{aligned} P_{XY} - P_{YX} &= -st(\partial_\mu\Gamma_\nu - \partial_\nu\Gamma_\mu + \Gamma_\mu\Gamma_\nu - \Gamma_\nu\Gamma_\mu) \\ &\quad + O(|st|(|s| + |t|)). \end{aligned}$$

The remainder contains both side lengths because the exact difference vanishes whenever $s = 0$ or $t = 0$. The matrix in parentheses is the curvature matrix in (2.3). Finally,

$$H_{s,t} - I = P_{YX}^{-1}(P_{XY} - P_{YX}), \quad P_{YX}^{-1} = I + O(|s| + |t|).$$

Multiplying by the return map therefore changes only the remainder. Returning to the starting tangent space gives

$$H_{s,t}v - v = -st R_p(X_p, Y_p)v + O(|st|(|s| + |t|)). \quad (2.5)$$

The remainder is measured in any fixed norm on the finite-dimensional space $T_p M$. The expansion concerns fixed coordinate directions as $s, t \rightarrow 0$.

The minus sign follows from the traversal $+X, +Y, -X, -Y$ and definition (2.1). Reversing the loop reverses its leading change. The factor st is oriented coordinate area. For a Riemannian metric, the leading physical area magnitude is

$$|st|\sqrt{g_p(X_p, X_p)g_p(Y_p, Y_p) - g_p(X_p, Y_p)^2}.$$

This conversion uses the metric on the two-plane spanned by the sides.

2.5 Coordinates and local flatness

The normal-coordinate construction in GR1 sets the Levi-Civita coefficients to zero at any chosen point p . In such coordinates, (2.3) becomes

$$R^\rho{}_{\sigma\mu\nu}(p) = \partial_\mu \Gamma^\rho{}_{\nu\sigma}(p) - \partial_\nu \Gamma^\rho{}_{\mu\sigma}(p).$$

The first derivatives of the connection can remain nonzero. Choosing a frame in which the connection vanishes at one point consequently does not remove the change produced by a small loop through nearby points. A loop samples how the connection varies across a region.

If instead all connection coefficients vanish throughout an open coordinate region, their derivatives vanish there as well, and the curvature is zero. For a Levi-Civita connection the converse holds locally. Where the curvature vanishes, sufficiently small neighborhoods admit coordinates with constant metric coefficients. Such a metric is called *locally flat*. The qualifier refers to a neighborhood, whereas the condition $R_p = 0$ at one isolated point says nothing about curvature at its neighbors.

These statements concern local geometry. A vanishing curvature tensor does not by itself determine the global topology or identify all paths between distant points. In the small-loop construction, a single coordinate neighborhood is sufficient. This is also why curvature is a local tensor, while holonomy may contain information obtained by transport around much larger loops.

3 Flat and curved spherical geometry

SOURCE SLIDES 10–11

3.1 Euclidean space in spherical coordinates

A nonzero connection coefficient can arise because the coordinate basis changes from point to point. We can test whether this produces curvature by calculating R . Consider Euclidean $\mathbb{R}^3$, with

$$x = r \sin \vartheta \cos \varphi, \quad y = r \sin \vartheta \sin \varphi, \quad z = r \cos \vartheta.$$

Work on a chart with $r > 0$, $0 < \vartheta < \pi$, and φ in an interval avoiding a longitude cut. The coordinate vectors have lengths $1, r, r \sin \vartheta$. They are related to the familiar unit vectors by

$$\partial_r = \hat{e}_r, \quad \partial_\vartheta = r \hat{e}_\vartheta, \quad \partial_\varphi = r \sin \vartheta \hat{e}_\varphi.$$

Their scalar products give

$$g = dr \otimes dr + r^2 d\vartheta \otimes d\vartheta + r^2 \sin^2 \vartheta d\varphi \otimes d\varphi. \quad (3.1)$$

The inverse metric is diagonal, with entries $1, r^{-2}, r^{-2} \sin^{-2} \vartheta$.

Applying the Christoffel formula gives the nonzero coefficients

$$\begin{aligned}\Gamma^r_{\vartheta\vartheta} &= -r, & \Gamma^r_{\varphi\varphi} &= -r \sin^2 \vartheta, \\ \Gamma^\vartheta_{r\vartheta} &= \Gamma^\vartheta_{\vartheta r} = \frac{1}{r}, & \Gamma^\varphi_{r\varphi} &= \Gamma^\varphi_{\varphi r} = \frac{1}{r}, \\ \Gamma^\vartheta_{\varphi\varphi} &= -\sin \vartheta \cos \vartheta, & \Gamma^\varphi_{\vartheta\varphi} &= \Gamma^\varphi_{\varphi\vartheta} = \cot \vartheta.\end{aligned}\tag{3.2}$$

For example,

$$\Gamma^r_{\vartheta\vartheta} = -\frac{1}{2}\partial_r(r^2) = -r, \quad \Gamma^\vartheta_{r\vartheta} = \frac{1}{2}r^{-2}\partial_r(r^2) = \frac{1}{r}.$$

The factors arise from the changing scale of the angular coordinate vectors as the radius varies.

A useful diagnostic component of curvature is

$$\begin{aligned}R^\vartheta_{\varphi\vartheta\varphi} &= \partial_\vartheta\Gamma^\vartheta_{\varphi\varphi} - \partial_\varphi\Gamma^\vartheta_{\vartheta\varphi} + \Gamma^\vartheta_{\vartheta\lambda}\Gamma^\lambda_{\varphi\varphi} - \Gamma^\vartheta_{\varphi\lambda}\Gamma^\lambda_{\vartheta\varphi} \\ &= \partial_\vartheta(-\sin \vartheta \cos \vartheta) - 0 + \Gamma^\vartheta_{\vartheta r}\Gamma^r_{\varphi\varphi} - \Gamma^\vartheta_{\varphi r}\Gamma^r_{\vartheta\varphi} \\ &= (-\cos^2 \vartheta + \sin^2 \vartheta) + \frac{1}{r}(-r \sin^2 \vartheta) - (-\sin \vartheta \cos \vartheta) \cot \vartheta \\ &= (-\cos^2 \vartheta + \sin^2 \vartheta) - \sin^2 \vartheta + \cos^2 \vartheta = 0.\end{aligned}$$

Only $\lambda = r$ survives in the first product and only $\lambda = \varphi$ in the second. The radial contribution $\Gamma^\vartheta_{\vartheta r}\Gamma^r_{\varphi\varphi} = -\sin^2 \vartheta$ will distinguish Euclidean space from the sphere.

The calculation illustrates flatness. It is not necessary to check every component separately. In Cartesian coordinates the metric is constant, all Christoffel symbols vanish, and (2.3) gives $R = 0$. Since curvature is a tensor, it vanishes in the spherical chart as well. The singular coefficients at the excluded axes concern the chart, while Euclidean geometry is smooth there.

Student work – flat space in a curved coordinate grid

Use (3.2) to calculate $R^r_{\vartheta r\vartheta}$. Identify the terms that cancel.

3.2 The round two-sphere

SOURCE SLIDE 12

Let $S_a^2 \subset \mathbb{R}^3$ be the sphere of fixed radius $a > 0$. Its angular coordinates and induced metric are

$$\begin{aligned}(x, y, z) &= a(\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta), \\ g &= a^2 d\vartheta \otimes d\vartheta + a^2 \sin^2 \vartheta d\varphi \otimes d\varphi.\end{aligned}\tag{3.3}$$

There is no radial tangent direction on this manifold. The connection coefficients intrinsic to the sphere are

$$\Gamma^\vartheta_{\varphi\varphi} = -\sin \vartheta \cos \vartheta, \quad \Gamma^\varphi_{\vartheta\varphi} = \Gamma^\varphi_{\varphi\vartheta} = \cot \vartheta,\tag{3.4}$$

with all others zero. They agree with the unit-sphere coefficients in GR1. Multiplying a metric by the constant a^2 multiplies its inverse by a^{-2} , so these factors cancel in the Christoffel formula.

The curvature calculation now sums only over ϑ, φ . The radial contribution found above is absent, giving

$$\begin{aligned} R^\vartheta_{\varphi\vartheta\varphi} &= \partial_\vartheta(-\sin\vartheta\cos\vartheta) - (-\sin\vartheta\cos\vartheta)\cot\vartheta \\ &= \sin^2\vartheta, \end{aligned} \quad (3.5)$$

$$R^\varphi_{\vartheta\vartheta\varphi} = \partial_\vartheta(\cot\vartheta) + \cot^2\vartheta = -1. \quad (3.6)$$

Exchanging the final two indices changes the sign. Components with equal final indices vanish. Direct substitution also gives $R^\vartheta_{\vartheta\vartheta\varphi} = R^\varphi_{\varphi\vartheta\varphi} = 0$, so these relations determine all components in this chart.

Curvature is therefore nonzero even though the sphere sits inside flat Euclidean space. Parallel transport on the sphere must keep a vector tangent to the sphere. The resulting connection is different from unrestricted transport in the ambient space.

3.3 Radius and infinitesimal transport

The coordinate components in (3.5)–(3.6) contain no a . Their physical interpretation requires the scale of the coordinate basis. Introduce the orthonormal frame

$$e_1 = \frac{1}{a}\partial_\vartheta, \quad e_2 = \frac{1}{a\sin\vartheta}\partial_\varphi.$$

The three input vectors each supply a factor a^{-1} . To express the output in the orthonormal frame, use $\partial_\vartheta = ae_1$ and $\partial_\varphi = a\sin\vartheta e_2$, supplying one factor a . Thus the radius dependence is $a^{-3}a = a^{-2}$. Substitution of the angular factors and the calculated components gives

$$R(e_1, e_2)e_1 = -\frac{1}{a^2}e_2, \quad R(e_1, e_2)e_2 = \frac{1}{a^2}e_1. \quad (3.7)$$

Thus the curvature measured using unit tangent vectors scales as a^{-2} . A larger sphere produces a smaller transport change around a loop of the same small physical area.

To make this precise, orient the coordinate rectangle by traversing $+\vartheta, +\varphi, -\vartheta, -\varphi$. Its leading physical area at p is

$$A = a^2 \sin\vartheta \Delta\vartheta \Delta\varphi.$$

Combining (2.5) with (3.7) shows that an initial vector e_1 returns with leading change

$$\Delta e_1 = \frac{A}{a^2}e_2.$$

The infinitesimal rotation is therefore A/a^2 in the positive orientation of the frame. This is the local curvature underlying the finite latitude transport studied in GR1. A finite loop generally requires integrating the transport equation along its path.

Student work – curvature and scale

Derive (3.7) from the coordinate components. Determine the leading change of $v = v^1 e_1 + v^2 e_2$ around the oriented small rectangle.

4 The algebraic structure of curvature

SOURCE SLIDE 13

The curvature tensor contains more information than a single measure of bending. Its components describe how parallel transport acts on each tangent direction when the path is varied in each coordinate plane. For the Levi–Civita connection, the metric and the vanishing of torsion impose identities that substantially reduce the number of independent components. Throughout this section, ∇ is the Levi–Civita connection of an n -dimensional pseudo-Riemannian manifold.

4.1 Curvature symmetries and the first Bianchi identity

Lower the first index with the metric and retain the index order used in the curvature definition,

$$R_{ijkl} = g_{ia} R^a{}_{jkl} = g(\partial_i, R(\partial_k, \partial_l)\partial_j). \quad (4.1)$$

The final two indices specify the two directions entering the curvature operator. The second index specifies the vector on which it acts, and the first pairs its output with another vector.

Proposition 4.1: Symmetries of Levi–Civita curvature

The lowered curvature tensor satisfies

$$R_{ijkl} = -R_{ijlk}, \quad R_{ijkl} = -R_{jikl}, \quad (4.2)$$

$$R_{ijkl} + R_{iklj} + R_{iljk} = 0, \quad (4.3)$$

$$R_{ijkl} = R_{klij}. \quad (4.4)$$

Equation (4.3) is the *first Bianchi identity*. It is also called the algebraic Bianchi identity.

The first antisymmetry follows directly from $R(X, Y) = -R(Y, X)$. For the second, apply the curvature commutator to the scalar $g(Z, W)$. Metric compatibility and the product rule give

$$g(R(X, Y)Z, W) + g(Z, R(X, Y)W) = 0.$$

Thus each operator $R(X, Y)$ is skew with respect to g , which is exactly antisymmetry in the first two lowered indices.

The first Bianchi identity in operator form is

$$R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0. \quad (4.5)$$

One way to verify it is to use the coordinate curvature formula. In the cyclic sum $R^i_{jkl} + R^i_{klj} + R^i_{ljk}$, the derivative terms cancel because $\Gamma^i_{jk} = \Gamma^i_{kj}$. The products of connection coefficients cancel in the same way. Since the sum is tensorial, the result holds for arbitrary vector fields. Pair interchange in (4.4) then follows algebraically from this cyclic identity and the two antisymmetries.

These assumptions matter. Antisymmetry in the final pair holds for any connection. Antisymmetry in the first pair uses metric compatibility, and the cyclic identity in the form above uses zero torsion. A general connection need not have all these symmetries.

Student work – interchanging the index pairs

Derive $R_{ijkl} = R_{klij}$ from the two antisymmetries and the first Bianchi identity.

4.2 The Ricci tensor and scalar curvature

Contract the output index with the first curvature direction,

$$R_{jl} = R^i_{jil} = g^{ik} R_{ijkl}, \quad R = g^{jl} R_{jl}. \quad (4.6)$$

The tensor $\text{Ric} = R_{jl} dx^j \otimes dx^l$ is the *Ricci tensor*, and the function $R : M \rightarrow \mathbb{R}$ is the *scalar curvature*. The symbol R denotes either the curvature operator or its scalar contraction according to its arguments and indices.

Pair interchange makes Ricci symmetric. Indeed,

$$R_{lj} = g^{ik} R_{ilkj} = g^{ik} R_{kijl} = g^{ik} R_{ijkl} = R_{jl},$$

where the final equality renames the summed indices. A contraction removes information. The Ricci tensor collects particular sums of curvature components, and the scalar curvature takes one further trace. Neither operation can generally be reversed.

4.3 The differential Bianchi identity

The curvature tensor also satisfies a differential identity,

$$\nabla_m R^i_{jkl} + \nabla_k R^i_{jlm} + \nabla_l R^i_{jmk} = 0. \quad (4.7)$$

This is the *second Bianchi identity*. Here $\nabla_m R$ means the covariant derivative of the complete curvature tensor, including a connection term for each of its four indices.

To prove it, fix an arbitrary point p and use coordinates in which $\Gamma^i_{jk}(p) = 0$, as constructed earlier. At p , the connection terms in ∇R vanish. In differentiating the coordinate formula for curvature, the derivatives of the quadratic terms $\Gamma\Gamma$ also vanish there. Consequently,

$$(\nabla_m R^i_{jkl})_p = (\partial_m \partial_k \Gamma^i_{lj} - \partial_m \partial_l \Gamma^i_{kj})_p.$$

The cyclic sum over m, k, l cancels because mixed partial derivatives commute. This proves the identity at p , and its tensorial character proves it in every chart and at every point.

A useful consequence follows by contraction. Set $i = k$ in (4.7) and sum. The first term contains $R^k_{jkl} = R_{jl}$. Antisymmetry in the final pair gives $R^k_{jmk} = -R^k_{jkm} = -R_{jm}$ in the third term. Thus

$$\nabla_m R_{jl} + \nabla_k R^k_{jlm} - \nabla_l R_{jm} = 0,$$

or, isolating the derivative of the full curvature tensor,

$$\nabla_k R^k_{jlm} = \nabla_l R_{jm} - \nabla_m R_{jl}.$$

Contracting with g^{jm} , using $\nabla g = 0$ and the curvature symmetries, gives

$$\nabla^j R_{jl} = \frac{1}{2} \nabla_l R, \quad \nabla^j = g^{jm} \nabla_m. \quad (4.8)$$

Therefore the symmetric tensor

$$G_{jl} = R_{jl} - \frac{1}{2} R g_{jl} \quad (4.9)$$

satisfies $\nabla^j G_{jl} = 0$. It is called the *Einstein tensor*. This divergence identity is a geometric consequence of the Levi-Civita connection and does not require any equation of motion.

Student work – contracting the Bianchi identity

Starting from $\nabla_k R^k_{jlm} = \nabla_l R_{jm} - \nabla_m R_{jl}$, show explicitly that contraction with g^{jm} gives $2\nabla^j R_{jl} = \nabla_l R$.

4.4 Separating scalar, Ricci, and Weyl curvature

For $n \geq 3$, we can separate the curvature information recorded by Ricci from the information lost in the contraction. The Ricci tensor itself has two parts, its scalar trace and a traceless remainder. First define the traceless Ricci tensor

$$Q_{ij} = R_{ij} - \frac{R}{n} g_{ij}, \quad g^{ij} Q_{ij} = 0. \quad (4.10)$$

We seek three tensors with the curvature symmetries whose Ricci contractions are, respectively, zero, Q_{jl} , and $(R/n)g_{jl}$. The required decomposition is

$$\begin{aligned} R_{ijkl} = & C_{ijkl} + \frac{1}{n-2} (g_{ik} Q_{jl} - g_{il} Q_{jk} - g_{jk} Q_{il} + g_{jl} Q_{ik}) \\ & + \frac{R}{n(n-1)} (g_{ik} g_{jl} - g_{il} g_{jk}). \end{aligned} \quad (4.11)$$

The last term is determined entirely by the scalar curvature. The middle term is determined by the part of Ricci left after its trace has been removed. The remaining tensor C is the *Weyl curvature tensor*.

The coefficients are fixed by taking traces. Contraction of the last term with g^{ik} gives $(R/n)g_{jl}$. Contraction of the four-term expression in parentheses gives $(n-2)Q_{jl}$, so the middle term contributes precisely Q_{jl} . The remainder therefore has zero Ricci contraction,

$$g^{ik}C_{ijkl} = 0.$$

Each term has the curvature symmetries, so all other contractions of C also vanish. This explains in what sense Weyl curvature is the part invisible to the Ricci tensor.

Counting independent components. The symmetries make the amount of remaining information precise. Antisymmetry means that each index pair can be restricted to $i < j$, giving $N = n(n-1)/2$ possible pairs. Pair interchange then makes the curvature components a symmetric $N \times N$ array, with $N(N+1)/2$ independent entries before the first Bianchi identity is imposed. Once the two antisymmetries and pair interchange are imposed, a cyclic identity with a repeated index adds no further condition. Each set of four distinct indices imposes one further independent relation. Thus the number of independent components is

$$\frac{N(N+1)}{2} - \binom{n}{4} = \frac{n^2(n^2-1)}{12},$$

where $\binom{n}{4} = 0$ for $n < 4$. For example, in four dimensions,

$$N = \frac{4 \cdot 3}{2} = 6, \quad \frac{N(N+1)}{2} - \binom{4}{4} = 21 - 1 = 20.$$

The six antisymmetric index pairs first give a symmetric array with 21 entries. The first Bianchi identity removes one independent entry. This counts the algebraic possibilities at a single point. The differential Bianchi identity instead restricts how curvature can vary from point to point.

The role of dimension. We can now ask how much curvature information remains after the Ricci tensor is specified. The answer depends on dimension. In three dimensions the curvature tensor has six independent components, as does its symmetric Ricci tensor. The explicit Ricci and scalar terms in (4.11) construct a curvature tensor with any prescribed symmetric Ricci tensor. The Ricci contraction is therefore surjective between two six-dimensional spaces, so it is also injective. There is no independent Weyl part. Thus $C = 0$, and Ricci determines the full curvature tensor.

In two dimensions formula (4.11) is unavailable because it divides by $n-2$. Instead there is only one independent curvature component, and

$$R_{ijkl} = \frac{R}{2}(g_{ik}g_{jl} - g_{il}g_{jk}), \quad R_{ij} = \frac{R}{2}g_{ij}. \quad (4.12)$$

These are pointwise statements about curvature. They do not imply that curvature at one point determines an entire manifold.

In four dimensions the Riemann tensor has 20 independent components at a point. A symmetric Ricci tensor has $4 \cdot 5/2 = 10$ components. Removing its one scalar trace leaves nine. The curvature decomposition therefore gives

$$\underbrace{20}_{\text{Riemann}} = \underbrace{1}_{\text{scalar}} + \underbrace{9}_{\text{traceless Ricci}} + \underbrace{10}_{\text{Weyl}}.$$

A vanishing Ricci tensor removes the scalar and traceless Ricci parts, but need not remove the Weyl part. Nonzero curvature can therefore remain in four dimensions.

Student work – traces of the decomposition

Contract the middle and final terms of (4.11) with g^{ik} . Verify their stated contributions to R_{jl} .

Student work – Ricci-flatness and dimension

What does $R_{ij} = 0$ imply about the full curvature tensor in three dimensions? What can remain in four dimensions?

5 Intrinsic curvature and bending in an ambient space

SOURCE SLIDE 14

The metric determines its Levi-Civita connection and hence its curvature without reference to a surrounding space. This is *intrinsic curvature*. If a manifold is embedded in another space, one can also study how its tangent spaces turn within that ambient space. Such bending is described by *extrinsic curvature*. The two notions measure different features, as the following examples show.

5.1 A circle and a cylinder

A circle of radius $a > 0$ in the Euclidean plane has metric

$$g = a^2 d\varphi \otimes d\varphi.$$

On an angular interval, the coordinate $s = a\varphi$ changes this to $g = ds \otimes ds$, the metric of a straight interval. Its intrinsic curvature is zero. More generally, in one dimension antisymmetry in the final two curvature indices forces the entire Riemann tensor to vanish. The circle nevertheless bends in the plane. Its unit tangent changes direction with arc length at rate $1/a$. That is extrinsic curvature. The local agreement of its metric with a line does not make the closed circle globally a line.

For a two-dimensional example, consider the circular cylinder parametrized locally by

$$F(\varphi, z) = (a \cos \varphi, a \sin \varphi, z), \quad a > 0.$$

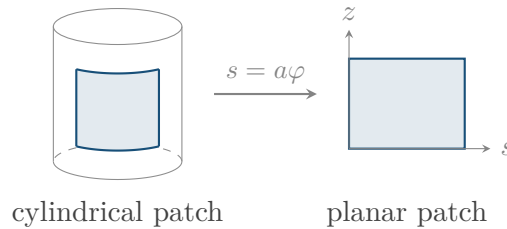
The Euclidean scalar products of

$$F_*\partial_\varphi = (-a \sin \varphi, a \cos \varphi, 0), \quad F_*\partial_z = (0, 0, 1)$$

give the induced metric

$$g = a^2 d\varphi \otimes d\varphi + dz \otimes dz.$$

All its coefficients are constant, so its Christoffel symbols and curvature vanish in this chart. Equivalently, setting $s = a\varphi$ gives the Euclidean metric $ds^2 + dz^2$. Unrolling a cylindrical patch preserves its intrinsic lengths and angles. The cylinder bends in $\mathbb{R}^3$, but is intrinsically flat.



5.2 Sectional curvature and the sphere

Curvature can be assigned to a tangent two-plane rather than averaged over all directions. Let $u, v \in T_p M$ span a plane on which the metric is nondegenerate. Thus $g(u, u)g(v, v) - g(u, v)^2 \neq 0$. Its *sectional curvature* is

$$K(u, v) = \frac{g(R(u, v)v, u)}{g(u, u)g(v, v) - g(u, v)^2}. \quad (5.1)$$

Both numerator and denominator acquire the same squared determinant under a change of basis of the plane, so the quotient depends only on the plane. For a Riemannian metric, every two-plane satisfies the nondegeneracy condition. In two dimensions there is only one tangent two-plane at each point. Its sectional curvature is the *Gaussian curvature*, and (4.12) gives $R = 2K$.

For the sphere of radius a ,

$$g = a^2 d\vartheta^2 + a^2 \sin^2 \vartheta d\varphi^2.$$

The previously calculated component $R^\vartheta_{\varphi\vartheta\varphi} = \sin^2 \vartheta$ becomes

$$R_{\vartheta\varphi\vartheta\varphi} = a^2 \sin^2 \vartheta.$$

Taking $u = \partial_\vartheta$ and $v = \partial_\varphi$ away from the coordinate poles therefore yields

$$K = \frac{a^2 \sin^2 \vartheta}{a^4 \sin^2 \vartheta} = \frac{1}{a^2}, \quad R = \frac{2}{a^2}. \quad (5.2)$$

These are scalar quantities and extend smoothly over the poles. The Ricci contraction can also be evaluated directly. Reversing both antisymmetric index pairs gives $R_{\varphi\vartheta\varphi\vartheta} = R_{\vartheta\varphi\vartheta\varphi}$, so

$$R_{\vartheta\vartheta} = g^{\varphi\varphi} R_{\varphi\vartheta\varphi\vartheta} = \frac{a^2 \sin^2 \vartheta}{a^2 \sin^2 \vartheta} = 1.$$

The corresponding contraction gives $R_{\varphi\varphi} = \sin^2 \vartheta$, and the mixed components vanish. Hence $\text{Ric} = a^{-2}g$. Taking its trace recovers the scalar curvature,

$$R = \frac{1}{a^2} 1 + \frac{1}{a^2 \sin^2 \vartheta} \sin^2 \vartheta = \frac{2}{a^2}.$$

If every nondegenerate tangent two-plane has the same sectional curvature K , the curvature tensor takes the form

$$R_{ijkl} = K(g_{ik}g_{jl} - g_{il}g_{jk}), \quad R_{ij} = (n-1)Kg_{ij}, \quad R = n(n-1)K.$$

The sphere illustrates this special case. Its scalar curvature determines the radius because its geometry is already known to be spherical. On a general manifold, sectional curvatures can differ between planes. A single scalar cannot record those differences, and $R = 0$ does not generally imply flatness. Scalar curvature has dimensions of inverse length squared, but is not a universal local curvature radius.

For a Riemannian metric this loss of information can be seen directly. Choose an orthonormal basis $e_1, \dots, e_n$ at a point. The denominator of (5.1) is then one for each pair of distinct basis vectors, so

$$R = \sum_{i,j} g(R(e_i, e_j)e_j, e_i) = 2 \sum_{i < j} K(e_i, e_j).$$

The zero terms with $i = j$ drop out, and each remaining plane appears twice in the first sum. Scalar curvature therefore sums the sectional curvatures of the basis planes. Positive and negative contributions can cancel, and different distributions can produce the same sum. The sum is independent of the chosen orthonormal basis even though the individual planes change.

Student work – curvature under a constant rescaling

Let $\tilde{g} = b^2 g$ with constant $b > 0$. Determine how Γ^i_{jk} , R^i_{jkl} , R_{ij} , and the scalar curvature change. Check the result for a sphere.

6 Orthonormal frames and tetrads

SOURCE SLIDES 15–16

The coordinate calculations express geometry in terms of the functions $g_{\mu\nu}$ and $\Gamma^\rho_{\mu\nu}$. An orthonormal frame gives another description of the same geometry. Its metric components are constant, while the variation of the frame is carried by its connection coefficients. This is the local Lorentz-frame construction of GR1, now developed into a method for calculating curvature.

Work on an open set U of a four-dimensional Lorentzian manifold. Choose smooth vector fields e_a , with $a = 0, 1, 2, 3$, satisfying $g(e_a, e_b) = \eta_{ab}$, where $\eta = \text{diag}(1, -1, -1, -1)$. Write the frame and its dual coframe as

$$e_a = e_a^\mu \partial_\mu, \quad \varepsilon^a = \varepsilon^a_\mu dx^\mu, \quad \varepsilon^a(e_b) = \delta^a_b. \quad (6.1)$$

The coefficient matrices are inverse to one another. Consequently,

$$\varepsilon^a_\mu e_b^\mu = \delta^a_b, \quad e_a^\mu \varepsilon^a_\nu = \delta^\mu_\nu. \quad (6.2)$$

The frame is called a *tetrad* or *vierbein*. These terms are also used for its coefficient matrix or the inverse coframe matrix. Here the notation always distinguishes the two matrices.

The metric and its inverse have the representations

$$g = \eta_{ab} \varepsilon^a \otimes \varepsilon^b, \quad g_{\mu\nu} = \eta_{ab} \varepsilon^a_\mu \varepsilon^b_\nu, \quad (6.3)$$

$$g^{\mu\nu} = \eta^{ab} e_a^\mu e_b^\nu. \quad (6.4)$$

Greek indices refer to the coordinate basis and Latin indices to the orthonormal frame. The metrics $g_{\mu\nu}$ and η_{ab} raise and lower indices of their respective kinds. The tetrad converts between the two kinds. For a vector V and a covector A ,

$$V^a = \varepsilon^a_\mu V^\mu, \quad V^\mu = e_a^\mu V^a, \quad A_a = e_a^\mu A_\mu. \quad (6.5)$$

For a mixed tensor, the same rule applies to each index separately,

$$Q^a_b = \varepsilon^a_\mu e_b^\nu Q^\mu_\nu.$$

Thus a tetrad changes the representation of a tensor without changing the tensor itself.

The distinction between index conversion and index lowering is useful here. Lowering V^μ produces the components of the metric-dual covector. Multiplying by ε^a_μ instead produces the components of the original vector in another basis. Accordingly, e_a^μ and ε^a_μ are inverse matrices, not arrays obtained from one another merely by changing the placement of an index. The metric relates them through

$$\varepsilon^a_\mu = \eta^{ab} g_{\mu\nu} e_b^\nu.$$

This follows by taking the scalar product of e_b with $\partial_\mu = \varepsilon^c_\mu e_c$.

An orthonormal frame need not be a coordinate frame. Its vectors may have nonzero Lie brackets, and its coframe forms need not be exact. For instance, the orthonormal coframe on the sphere in GR1 satisfies $d\varepsilon^2 \neq 0$. Constant metric components in such a frame therefore do not imply zero curvature. All constructions here are local. A single smooth orthonormal frame on the entire manifold is not assumed.

Student work – constant metric components

Why can $g(e_a, e_b) = \eta_{ab}$ hold throughout a neighborhood with nonzero curvature? Explain the difference from constant metric components in a coordinate basis.

6.1 Coordinate changes and local Lorentz transformations

Under a coordinate change $x^\mu \mapsto x'^\nu$, keep the geometric frame fixed. Its coefficients transform as

$$e_a{}^{\nu'} = \frac{\partial x'^{\nu'}}{\partial x^\mu} e_a^\mu, \quad \varepsilon^a{}_{\nu'} = \frac{\partial x^\mu}{\partial x'^{\nu'}} \varepsilon^a_\mu.$$

These are the usual transformations of a vector and a covector. Alternatively, keep the coordinates fixed and choose a different orthonormal frame. Following the component convention of SR1 and GR1, write

$$e'_a = e_b (\Lambda^{-1})^b{}_a, \quad \Lambda^\top \eta \Lambda = \eta. \quad (6.6)$$

Here $\Lambda : U \rightarrow O(1, 3)$ is a smooth matrix-valued function. Duality and the equality $V = V^a e_a = V'^a e'_a$ then give

$$\varepsilon'^a = \Lambda^a_b \varepsilon^b, \quad V'^a = \Lambda^a_b V^b, \quad A'_a = (\Lambda^{-1})^b_a A_b. \quad (6.7)$$

The word *local* allows the Lorentz matrix to vary with position. Equation (6.6) preserves (6.3), so both frames describe the same metric.

Coordinate changes act on the Greek indices, while frame changes act on the Latin indices. Both freedoms remain available. A tetrad has sixteen component functions, with six local Lorentz freedoms. This leaves the ten independent functions of a symmetric metric. The tetrad description reorganizes the metric information together with a choice of frame.

For an observer with four-velocity u normalized by $g(u, u) = c^2$, choose $e_0 = u/c$ at that point. The three spacelike frame vectors then provide orthogonal spatial directions for that observer. This interpretation concerns a basis of one tangent space. Extending the basis smoothly over a neighborhood provides a family of local frames, and does not by itself provide coordinates on that neighborhood.

Student work – converting tensor components

Express $Q^\mu{}_\nu$ in terms of $Q^a{}_b$. Show that $Q^a{}_a = Q^\mu{}_\mu$.

7 The connection in an orthonormal frame

SOURCE SLIDE 17

Let ∇ be an affine connection. Define its coefficients in the orthonormal frame by

$$\nabla_{\partial_\mu} e_b = \omega_\mu{}^a{}_b e_a, \quad \omega^a{}_b = \omega_\mu{}^a{}_b dx^\mu. \quad (7.1)$$

The matrix $\omega = (\omega^a{}_b)$ is the connection one-form in this frame. For a metric-compatible connection it is commonly called the *spin connection*. Here its action is needed only on vectors and tensors.

Differentiating $V = V^b e_b$ gives

$$(\nabla_\mu V)^a = \partial_\mu V^a + \omega_\mu{}^a{}_b V^b. \quad (7.2)$$

Preservation of the pairing $A_a V^a$ determines the sign for a lower frame index. Thus

$$(\nabla_\mu A)_a = \partial_\mu A_a - \omega_\mu{}^b{}_a A_b, \quad (7.3)$$

$$(\nabla_\mu Q)^a{}_b = \partial_\mu Q^a{}_b + \omega_\mu{}^a{}_c Q^c{}_b - \omega_\mu{}^c{}_b Q^a{}_c. \quad (7.4)$$

These are the tensor differentiation rules already obtained for coordinate indices, now applied to the moving orthonormal frame.

7.1 Relating the two connection matrices

The coefficients $\Gamma^\rho_{\mu\nu}$ and $\omega_\mu{}^a{}_b$ describe the same derivative in different frames. To find their relation, substitute $V^a = \varepsilon^a{}_\nu V^\nu$ into (7.2). Expressing the answer in the coordinate basis gives

$$\begin{aligned} e_a{}^\rho (\nabla_\mu V)^a &= e_a{}^\rho \left[\partial_\mu (\varepsilon^a{}_\nu V^\nu) + \omega_\mu{}^a{}_b \varepsilon^b{}_\nu V^\nu \right] \\ &= e_a{}^\rho \varepsilon^a{}_\nu \partial_\mu V^\nu + e_a{}^\rho (\partial_\mu \varepsilon^a{}_\nu) V^\nu + e_a{}^\rho \omega_\mu{}^a{}_b \varepsilon^b{}_\nu V^\nu \\ &= \partial_\mu V^\rho + e_a{}^\rho \left(\partial_\mu \varepsilon^a{}_\nu + \omega_\mu{}^a{}_b \varepsilon^b{}_\nu \right) V^\nu. \end{aligned}$$

The first term reduces to $\partial_\mu V^\rho$ because $e_a{}^\rho \varepsilon^a{}_\nu = \delta^\rho_\nu$. Since the result equals $\partial_\mu V^\rho + \Gamma^\rho_{\mu\nu} V^\nu$ for every vector field,

$$\Gamma^\rho_{\mu\nu} = e_a{}^\rho \left(\partial_\mu \varepsilon^a{}_\nu + \omega_\mu{}^a{}_b \varepsilon^b{}_\nu \right). \quad (7.5)$$

Equivalently,

$$\omega_\mu{}^a{}_b = \varepsilon^a{}_\rho \left(\partial_\mu e_b{}^\rho + \Gamma^\rho_{\mu\nu} e_b{}^\nu \right). \quad (7.6)$$

The derivative of the frame is essential. A connection cannot be converted by multiplying tetrad matrices alone.

Multiplying (7.5) by $\varepsilon^a{}_\rho$ yields the *tetrad postulate*,

$$\mathcal{D}_\mu \varepsilon^a{}_\nu := \partial_\mu \varepsilon^a{}_\nu - \Gamma^\rho_{\mu\nu} \varepsilon^a{}_\rho + \omega_\mu{}^a{}_b \varepsilon^b{}_\nu = 0. \quad (7.7)$$

The symbol $\mathcal{D}_\mu$ indicates that both the coordinate index and the frame index are differentiated. This identity states that conversion of components commutes with covariant differentiation. Once ω is defined from ∇ and the frame, it is an identity, not an additional condition on the geometry. In particular, it holds for connections with torsion.

7.2 Transformation of the spin connection

SOURCE SLIDE 18

For a fixed orthonormal frame, $\omega^a{}_b$ is a one-form under coordinate changes. Its frame transformation contains an additional term because Λ can depend on position. Write the component column derivative as

$$DV = dV + \omega V.$$

It must satisfy $D'V' = \Lambda DV$. Using $V' = \Lambda V$,

$$D'V' = d(\Lambda V) + \omega' \Lambda V = \Lambda dV + (d\Lambda + \omega' \Lambda) V.$$

Comparison for arbitrary V determines

$$\boxed{\omega' = \Lambda \omega \Lambda^{-1} - d\Lambda \Lambda^{-1}.} \quad (7.8)$$

The last term compensates for differentiating the varying Lorentz matrix. It vanishes for a constant frame transformation. This is the frame version of the inhomogeneous coordinate transformation law for $\Gamma^\rho_{\mu\nu}$.

Student work – the inverse tetrad identity

Differentiate $\varepsilon^a_\rho e_b^\rho = \delta^a_b$ and use the tetrad postulate to derive (7.6).

8 Covariant exterior differentiation

SOURCE SLIDES 18–19

Differential forms keep antisymmetric coordinate indices together. They are therefore particularly useful for torsion and curvature, whose final two indices are antisymmetric. To differentiate forms that also carry frame indices, combine the exterior derivative with the connection acting on those indices.

A tangent-vector-valued k -form can be written $\alpha = e_a \otimes \alpha^a$, where each $\alpha^a \in \Omega^k(U)$ is an ordinary differential form. Its components transform as $\alpha'^a = \Lambda^a_b \alpha^b$. Define the *covariant exterior derivative* by

$$(D\alpha)^a = d\alpha^a + \omega^a_b \wedge \alpha^b. \quad (8.1)$$

The result is a vector-valued $(k+1)$ -form. Substitution of (7.8) shows that $(D\alpha)'^a = \Lambda^a_b (D\alpha)^b$. For a covector-valued form β_a , the corresponding rule is

$$(D\beta)_a = d\beta_a - \omega^b_a \wedge \beta_b.$$

For $k=0$, equation (8.1) recovers the vector derivative above.

Derivative notation

d acts on ordinary scalar-valued differential forms. It increases form degree by one and requires no connection.

D acts on forms that also carry vector or tensor indices. It increases form degree by one and adds connection terms for those indices, as in (8.1).

$\mathcal{D}_\mu$ is the full component covariant derivative. It includes a connection term for every tensor index, using Γ for coordinate indices and ω for frame indices. The new differentiation index μ is retained rather than antisymmetrized with the form indices.

For a vector-valued one-form $\alpha^a = \alpha^a_\nu dx^\nu$, the component convention is

$$(D\alpha)^a = \frac{1}{2} (D\alpha)^a_{\mu\nu} dx^\mu \wedge dx^\nu, \quad (8.2)$$

$$(D\alpha)^a_{\mu\nu} = \partial_\mu \alpha^a_\nu - \partial_\nu \alpha^a_\mu + \omega_\mu^a_b \alpha^b_\nu - \omega_\nu^a_b \alpha^b_\mu. \quad (8.3)$$

The factor $1/2$ counts each antisymmetric pair once. The exterior derivative already handles the form indices, so the displayed definition adds connection terms only for the frame index.

For comparison, the full derivative of the component array is

$$\mathcal{D}_\mu \alpha^a_\nu = \partial_\mu \alpha^a_\nu + \omega_\mu^a_b \alpha^b_\nu - \Gamma^\rho_{\mu\nu} \alpha^a_\rho.$$

Subtracting the expression with μ and ν exchanged gives

$$\mathcal{D}_\mu \alpha^a{}_\nu - \mathcal{D}_\nu \alpha^a{}_\mu = (D\alpha)^a{}_{\mu\nu} - (\Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu})\alpha^a{}_\rho.$$

The expression in parentheses is $T^\rho{}_{\mu\nu}$ in a coordinate basis. Moving this term to the other side gives

$$(D\alpha)^a{}_{\mu\nu} = \mathcal{D}_\mu \alpha^a{}_\nu - \mathcal{D}_\nu \alpha^a{}_\mu + T^\rho{}_{\mu\nu} \alpha^a{}_\rho.$$

For the Levi–Civita connection the torsion term vanishes, and covariant exterior differentiation is antisymmetrized covariant differentiation. The definition using d and ω remains valid for any connection. This distinction is especially useful when the one-form being differentiated is the coframe itself.

Matrix-valued forms require both matrix multiplication and the wedge product. Our notation means

$$(A \wedge B)^a{}_b = A^a{}_c \wedge B^c{}_b.$$

Their factors generally cannot be exchanged using a sign alone, because the matrix coefficients need not commute. For an endomorphism-valued k -form $A = (A^a{}_b)$, covariant exterior differentiation takes the form

$$DA = dA + \omega \wedge A - (-1)^k A \wedge \omega. \quad (8.4)$$

The final term differentiates the lower frame index. Its degree sign appears when the connection one-form is moved past the k -form. In particular, for a matrix of functions, $DA = dA + \omega A - A\omega$.

8.1 Torsion and the first structure equation

The collection ε^a represents the tangent-vector-valued one-form given by the identity map $X \mapsto X$, since $X = \varepsilon^a(X)e_a$. Its covariant exterior derivative measures torsion. To see this, evaluate it on vector fields X, Y ,

$$\begin{aligned} (D\varepsilon)^a(X, Y) &= X(\varepsilon^a(Y)) - Y(\varepsilon^a(X)) - \varepsilon^a([X, Y]) \\ &\quad + \omega^a{}_b(X)\varepsilon^b(Y) - \omega^a{}_b(Y)\varepsilon^b(X) \\ &= \varepsilon^a(\nabla_X Y - \nabla_Y X - [X, Y]). \end{aligned}$$

Thus the torsion two-forms satisfy the first Cartan structure equation,

$$\boxed{T^a = d\varepsilon^a + \omega^a{}_b \wedge \varepsilon^b}, \quad T^a = \frac{1}{2} \varepsilon^a{}_\rho T^\rho{}_{\mu\nu} dx^\mu \wedge dx^\nu. \quad (8.5)$$

For a torsion-free connection this becomes an equation for the connection one-forms in terms of the chosen coframe.

This equation also incorporates the noncommutativity of the frame. Write $[e_b, e_c] = C^a{}_{bc}e_a$. The coframe identity from GR1 is $d\varepsilon^a(e_b, e_c) = -C^a{}_{bc}$, so

$$T^a(e_b, e_c) = \omega^a{}_c(e_b) - \omega^a{}_b(e_c) - C^a{}_{bc}.$$

Thus zero torsion does not require the frame connection coefficients to be symmetric in the direction and vector indices. Their antisymmetric part compensates for the Lie bracket of the frame vectors. The familiar symmetry of the lower Christoffel indices is the coordinate-frame case, where all these brackets vanish. In the structure equation this distinction is contained in $d\varepsilon^a$, so no separate bracket correction needs to be added during the calculation.

The operators D and $\mathcal{D}_\mu$ in the tetrad postulate serve different purposes. The latter differentiates every component index. Antisymmetrizing the tetrad postulate gives

$$(D\varepsilon)^a{}_{\mu\nu} = \varepsilon^a{}_\rho (\Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu}),$$

which can be nonzero even though $\mathcal{D}_\mu \varepsilon^a{}_\nu = 0$.

8.2 Curvature and the second structure equation

Apply D twice to a vector-component column V . The ordinary exterior derivative satisfies $d^2 = 0$, but the connection adds terms,

$$\begin{aligned} D^2 V &= d(dV + \omega V) + \omega \wedge (dV + \omega V) \\ &= (d\omega)V - \omega \wedge dV + \omega \wedge dV + (\omega \wedge \omega)V \\ &= (d\omega + \omega \wedge \omega)V. \end{aligned}$$

This defines the curvature two-form and gives the second Cartan structure equation,

$$\boxed{\mathcal{R}^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b}, \quad D^2 V^a = \mathcal{R}^a{}_b V^b. \quad (8.6)$$

Evaluation on two vector fields gives the curvature operator,

$$\mathcal{R}^a{}_b(X, Y)V^b = \varepsilon^a(R(X, Y)V).$$

The coordinate and frame descriptions are therefore related by

$$\mathcal{R}^a{}_b = \frac{1}{2} \varepsilon^a{}_\rho e_b{}^\sigma R^\rho{}_{\sigma\mu\nu} dx^\mu \wedge dx^\nu = \frac{1}{2} R^a{}_{bcd} \varepsilon^c \wedge \varepsilon^d. \quad (8.7)$$

The two-form notation packages the antisymmetric direction indices. It does not change the curvature convention or add a new curvature tensor. Under a local Lorentz change,

$$T'^a = \Lambda^a{}_b T^b, \quad \mathcal{R}' = \Lambda \mathcal{R} \Lambda^{-1}.$$

The inhomogeneous terms in the connection transformation cancel from these tensorial quantities.

The same calculation also applies to a vector-valued form of any degree. The product rule gives $d(\omega \wedge \alpha) = d\omega \wedge \alpha - \omega \wedge d\alpha$, so the terms containing $d\alpha$ cancel and

$$D^2 \alpha^a = \mathcal{R}^a{}_b \wedge \alpha^b.$$

Thus D generally fails to square to zero. Its failure is measured precisely by curvature acting on the vector-valued part of the form. Ordinary scalar-valued forms have no frame index, and their ordinary exterior derivative still satisfies $d^2 = 0$.

Student work – curvature acting on covectors

Starting from $(DA)_a = dA_a - \omega^b{}_a A_b$, show that $D^2 A_a = -\mathcal{R}^b{}_a A_b$.

8.3 The Bianchi identities

Both Bianchi identities follow from the structure equations and the graded product rule. First, differentiate the torsion equation,

$$\begin{aligned} DT^a &= d(d\varepsilon^a + \omega^a_b \wedge \varepsilon^b) + \omega^a_b \wedge (d\varepsilon^b + \omega^b_c \wedge \varepsilon^c) \\ &= (d\omega^a_b) \wedge \varepsilon^b - \omega^a_b \wedge d\varepsilon^b + \omega^a_b \wedge d\varepsilon^b + \omega^a_b \wedge \omega^b_c \wedge \varepsilon^c. \end{aligned}$$

The middle terms cancel. Relabeling the summed frame indices gives

$$\boxed{DT^a = \mathcal{R}^a_b \wedge \varepsilon^b.} \quad (8.8)$$

When $T = 0$, the right-hand side vanishes. Evaluation on three vector fields yields $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$, the algebraic first Bianchi identity. Equation (8.8) also states its form for a connection with torsion.

For the curvature matrix, equation (8.4) with degree two gives

$$D\mathcal{R} = d\mathcal{R} + \omega \wedge \mathcal{R} - \mathcal{R} \wedge \omega.$$

The structure equation implies

$$\begin{aligned} d\mathcal{R} &= d\omega \wedge \omega - \omega \wedge d\omega, \\ \omega \wedge \mathcal{R} - \mathcal{R} \wedge \omega &= \omega \wedge d\omega - d\omega \wedge \omega. \end{aligned}$$

The cubic terms cancel by associativity of matrix multiplication and the wedge product. Hence the differential Bianchi identity is

$$\boxed{D\mathcal{R} = 0.} \quad (8.9)$$

For the Levi-Civita connection, its coordinate expression is

$$\nabla_\lambda R^\rho_{\sigma\mu\nu} + \nabla_\mu R^\rho_{\sigma\nu\lambda} + \nabla_\nu R^\rho_{\sigma\lambda\mu} = 0.$$

Thus the compact form equations reproduce the curvature identities used earlier, while making their origin in the connection explicit.

8.4 Metric compatibility in the orthonormal frame

The frame metric has constant entries. Metric compatibility therefore reduces to

$$0 = (\nabla_\mu \eta)_{ab} = -\omega_\mu^c{}_a \eta_{cb} - \omega_\mu^c{}_b \eta_{ac}.$$

Define $\omega_{ab} = \eta_{ac} \omega^c{}_b$. Then

$$\omega_{ab} = -\omega_{ba}. \quad (8.10)$$

The antisymmetry applies after lowering the first frame index. For a positive-definite orthonormal frame, η_{ab} is replaced by δ_{ab} , so the mixed-index matrix is antisymmetric as well. Together with $T^a = 0$, equation (8.10) determines the Levi-Civita connection in a chosen orthonormal frame. The following calculations use these two conditions directly.

For Lorentzian signature the location of the indices matters even for this simple condition. For a spatial index i , $\omega_{0i} = \omega^0{}_i$ and $\omega_{i0} = -\omega^i{}_0$, so antisymmetry implies $\omega^0{}_i = \omega^i{}_0$. The mixed matrix is therefore not an ordinary antisymmetric real matrix. It obeys $\omega^\top \eta + \eta \omega = 0$, which is the infinitesimal form of the Lorentz condition.

In practice, one first computes $d\varepsilon^a$, then solves $d\varepsilon^a + \omega^a_b \wedge \varepsilon^b = 0$ together with metric compatibility. These equations determine the connection uniquely by the Levi-Civita theorem. Finally, the second structure equation gives the curvature. This method avoids calculating all coordinate Christoffel symbols when the metric has a simple orthonormal coframe.

8.5 Flat space and the sphere revisited

Example 8.1: Euclidean space in a spherical coframe

On a spherical coordinate patch with $r > 0$ and $0 < \vartheta < \pi$, take

$$\varepsilon^1 = dr, \quad \varepsilon^2 = r d\vartheta, \quad \varepsilon^3 = r \sin \vartheta d\varphi.$$

The Euclidean metric is $g = \delta_{ab} \varepsilon^a \otimes \varepsilon^b$. The exterior derivatives are

$$\begin{aligned} d\varepsilon^1 &= 0, \\ d\varepsilon^2 &= \frac{1}{r} \varepsilon^1 \wedge \varepsilon^2, \\ d\varepsilon^3 &= (\sin \vartheta dr + r \cos \vartheta d\vartheta) \wedge d\varphi \\ &= \frac{1}{r} \varepsilon^1 \wedge \varepsilon^3 + \frac{\cot \vartheta}{r} \varepsilon^2 \wedge \varepsilon^3. \end{aligned}$$

These derivatives suggest trying $\omega^1_2 = A\varepsilon^2$, $\omega^1_3 = B\varepsilon^3$, and $\omega^2_3 = C\varepsilon^3$, with scalar coefficient functions A, B, C , together with antisymmetry. It is enough to find a solution of all three torsion equations, since metric compatibility and zero torsion determine the connection uniquely. The first torsion equation is then automatic. The second becomes

$$0 = d\varepsilon^2 + \omega^2_1 \wedge \varepsilon^1 + \omega^2_3 \wedge \varepsilon^3 = \left(\frac{1}{r} + A \right) \varepsilon^1 \wedge \varepsilon^2,$$

so $A = -1/r$. For the third equation, use $\omega^3_1 = -B\varepsilon^3$ and $\omega^3_2 = -C\varepsilon^3$. Then

$$\begin{aligned} 0 = T^3 &= d\varepsilon^3 + \omega^3_1 \wedge \varepsilon^1 + \omega^3_2 \wedge \varepsilon^2 \\ &= \frac{1}{r} \varepsilon^1 \wedge \varepsilon^3 + \frac{\cot \vartheta}{r} \varepsilon^2 \wedge \varepsilon^3 - B\varepsilon^3 \wedge \varepsilon^1 - C\varepsilon^3 \wedge \varepsilon^2 \\ &= \left(\frac{1}{r} + B \right) \varepsilon^1 \wedge \varepsilon^3 + \left(\frac{\cot \vartheta}{r} + C \right) \varepsilon^2 \wedge \varepsilon^3. \end{aligned}$$

The two basis two-forms are linearly independent. Their coefficients must vanish separately, so $B = -1/r$ and $C = -\cot \vartheta/r$. Substituting the coframe gives

$$\omega^1_2 = -d\vartheta, \quad \omega^1_3 = -\sin \vartheta d\varphi, \quad \omega^2_3 = -\cos \vartheta d\varphi,$$

with $\omega^b_a = -\omega^a_b$ and zero diagonal entries. For example, the angular curvature two-form is

$$\begin{aligned} \mathcal{R}^2_3 &= d\omega^2_3 + \omega^2_1 \wedge \omega^1_3 \\ &= \sin \vartheta d\vartheta \wedge d\varphi + (d\vartheta) \wedge (-\sin \vartheta d\varphi) = 0. \end{aligned}$$

The other curvature forms also vanish. The nonzero connection describes the variation of this frame in flat Euclidean space.

Student work – flatness in the spherical frame

For the connection above, calculate $\mathcal{R}^1_2$ and $\mathcal{R}^1_3$.

Example 8.2: Curvature of the sphere from its coframe

For the sphere of radius $a > 0$, use

$$\varepsilon^1 = a \, d\vartheta, \quad \varepsilon^2 = a \sin \vartheta \, d\varphi.$$

Here

$$d\varepsilon^1 = 0, \quad d\varepsilon^2 = \frac{\cot \vartheta}{a} \varepsilon^1 \wedge \varepsilon^2.$$

The torsion-free equations are

$$\omega^1_2 \wedge \varepsilon^2 = 0, \quad d\varepsilon^2 + \omega^2_1 \wedge \varepsilon^1 = 0.$$

Writing $\omega^1_2 = A\varepsilon^1 + B\varepsilon^2$, the first equation reads $A\varepsilon^1 \wedge \varepsilon^2 = 0$, so $A = 0$. Using $\omega^2_1 = -B\varepsilon^2$, the second becomes

$$0 = \frac{\cot \vartheta}{a} \varepsilon^1 \wedge \varepsilon^2 - B\varepsilon^2 \wedge \varepsilon^1 = \left(\frac{\cot \vartheta}{a} + B \right) \varepsilon^1 \wedge \varepsilon^2.$$

Thus $B = -\cot \vartheta/a$. Therefore

$$\omega^1_2 = -\cos \vartheta \, d\varphi, \quad \mathcal{R}^1_2 = d\omega^1_2 = \sin \vartheta \, d\vartheta \wedge d\varphi = \frac{1}{a^2} \varepsilon^1 \wedge \varepsilon^2.$$

There is no third frame direction and no quadratic connection term in this curvature component. Reading its coefficient gives $R^1_{212} = 1/a^2$, in agreement with the coordinate calculation. Thus the Gaussian curvature is $K = 1/a^2$ and the scalar curvature is $R = 2/a^2$.

The angular part of the Euclidean spherical connection resembles the sphere connection, but their curvature calculations include different frame directions. In Euclidean space, the terms involving the radial direction cancel the angular derivative term. On the sphere, the connection acts only within its two-dimensional tangent spaces, and its intrinsic curvature is nonzero.

Student work – Ricci curvature in the sphere frame

Use $\mathcal{R}^1_2 = a^{-2} \varepsilon^1 \wedge \varepsilon^2$ and curvature antisymmetry to find R_{11}, R_{22}, R_{12} . Contract them to obtain the scalar curvature.

9 Metric, connection, and curvature

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The metric determines scalar products and selects the Levi–Civita connection. This connection defines parallel transport, and curvature measures the leading change around a small loop. Nonzero connection coefficients can reflect a varying basis in flat space. Nonzero curvature is an intrinsic obstruction to a parallel frame throughout a neighborhood.

Coordinate bases and orthonormal frames give equivalent descriptions. The coframe records the metric, while the connection and curvature can be represented as follows.

$$\begin{aligned}\nabla_{\partial_\mu}\partial_\nu &= \Gamma^\rho_{\mu\nu}\partial_\rho, & \nabla e_b &= e_a \otimes \omega^a_b, \\ R(\partial_\mu, \partial_\nu)\partial_\sigma &= R^\rho_{\sigma\mu\nu}\partial_\rho, & \mathcal{R}^a_b &= d\omega^a_b + \omega^a_c \wedge \omega^c_b.\end{aligned}$$

Here the tensor-valued one-form ∇e_b sends X to $\nabla_X e_b$.

In the frame description, $T^a = 0$ and $\omega_{ab} = -\omega_{ba}$ together determine the Levi–Civita connection. Coordinate transformations remain available, and local Lorentz transformations express the freedom to choose an orthonormal frame at each event.